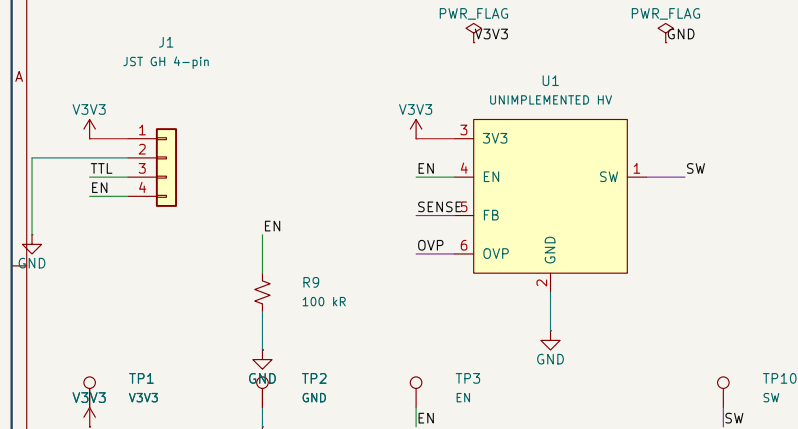
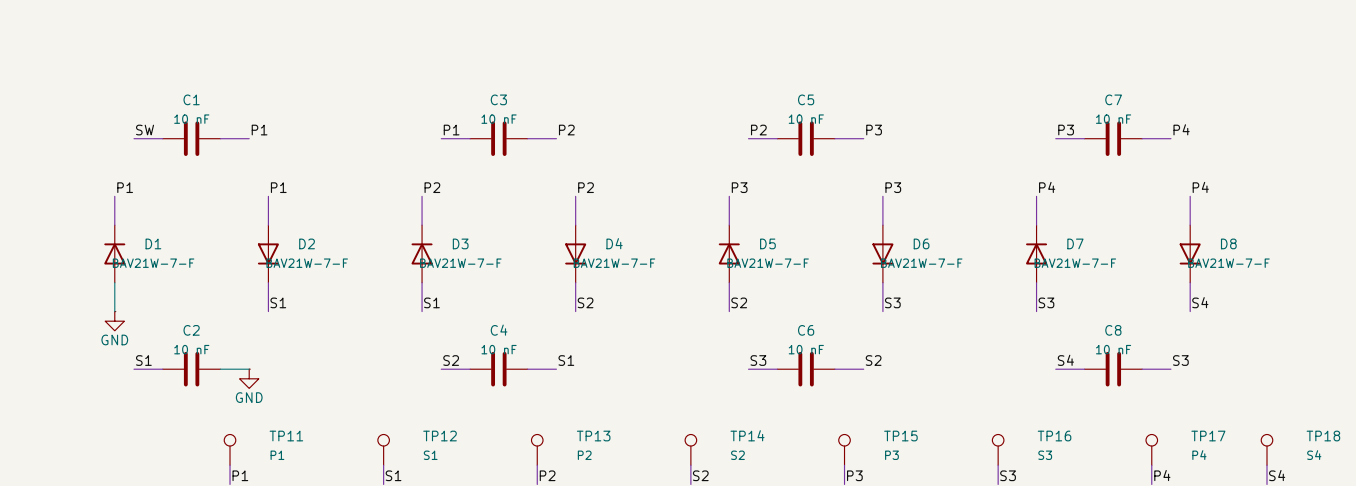


01 PI INPUT + ANALOG HV BOUNDARY



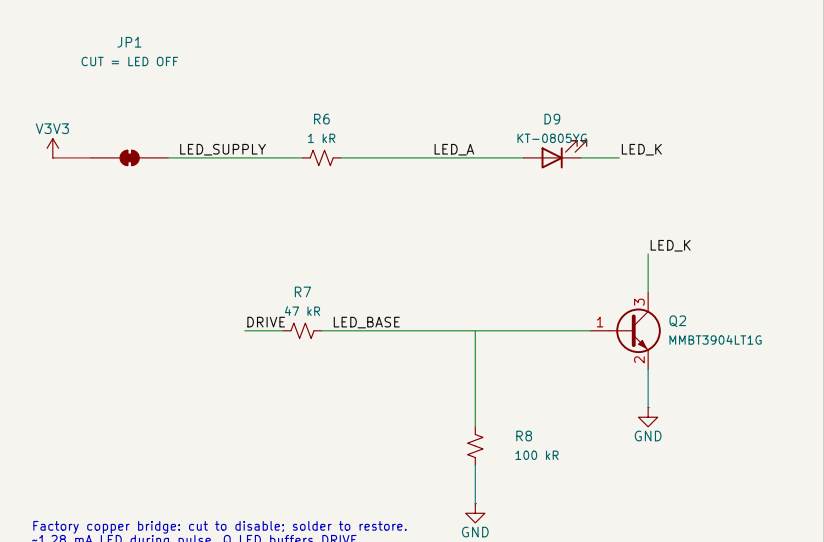
J1: 1=3V3 2=GND 3=PULSE 4=EN
 U1 is a functional boundary, not an IC pinout.
 1 mH; 10 kHz; 20 us ON; 22 uF modeled bypass.
 Implement oscillator / driver / feedback / soft-start.

02 FOUR-STAGE COCKCROFT-WALTON LADDER



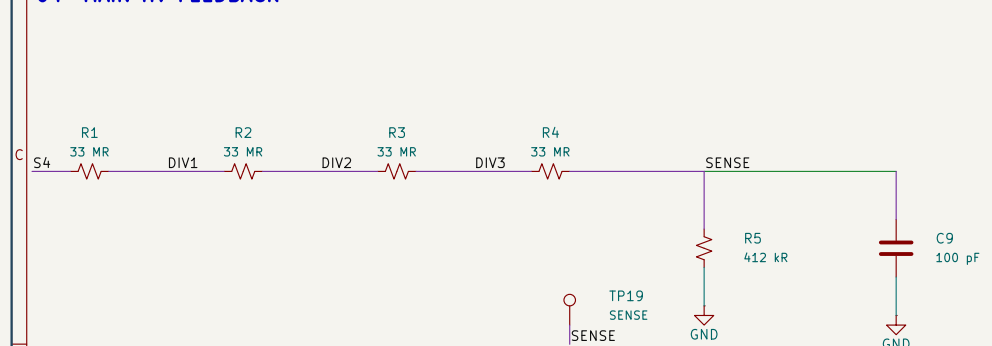
C1-C8: 10 nF COG / 630 V / 1206. D1-D8: BAV21W-7-F.
SW ~106 V peak; S4 ~399 V nominal. Verify diode reverse stress and leakage.

03 GREEN 0805 PULSE INDICATOR



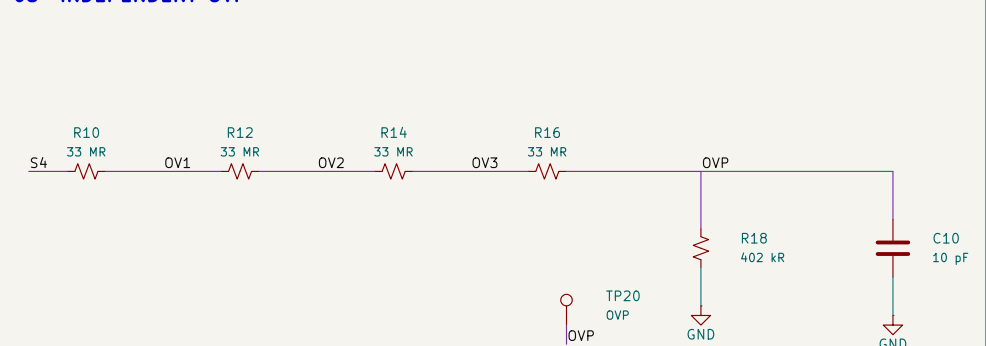
Factory copper bridge: cut to disable; solder to restore.
~1.28 mA LED during pulse. Q_LED buffers DRIVE.
No pulse stretching; flashes may be hard to see.

04 MAIN HV FEEDBACK



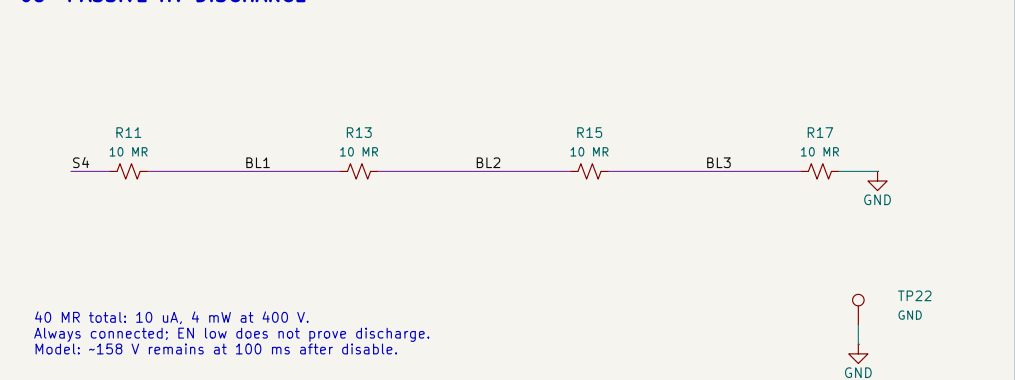
399.16 V = 1.242 x (132M + 412k) / 412k
3.02 uA at 400 V; C9 time constant ~41 us.

05 INDEPENDENT OVP



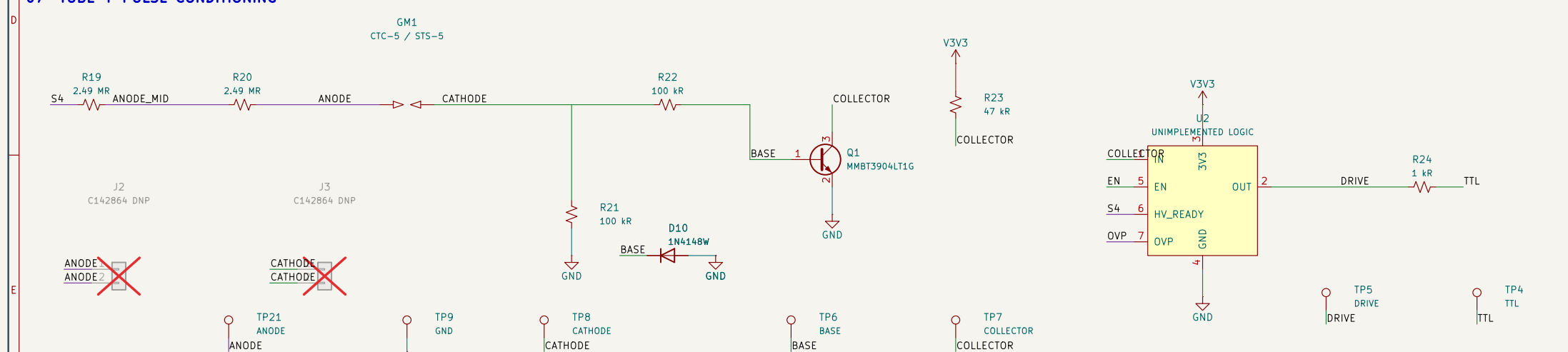
409.06 V = 1.242 x (132M + 402k) / 402k
Separate reference required; latch / inhibit driver.

06 PASSIVE HV DISCHARGE



40 MR total: 10 uA, 4 mW at 400 V.
Always connected; EN low does not prove discharge.
Model: ~158 V remains at 100 ms after disable.

07 TUBE + PULSE CONDITIONING



R_A1/R_A2: 2.49 MR EACH, ≥ 500 V working rating; ~ 88.4 μ A maximum at 440 V. GND
J2/J3: C142864 DNP for JLC, hand-fit clips. Tube supplied by user.
U2 inverts Q1 and blanks on EN / HV-ready / OVP. No MCU. Physical logic remains unfinished.
Plan 250 us tube dead time (reference ~ 190 us); dose conversion is approximate.

PROTOTYPE / HV SAFETY NOTES

400 V present on exposed pads / tube clips.
Discharge and verify before handling.
Do not connect ordinary Pi / scope probes to HV.

Preserved routing has unresolved clearances.
U1/U2 have no physical implementation.
This is NOT a fabrication release.

22 plated probe holes; 1 mm drill / 2 mm pads.
See README for model limits and DRC report.

U1/U2 unfinished; preserved PCB requires clearance review

muonTelescope

Sheet: /
File: glowcost-geiger.kicad_sch

Title: Glowcost Geiger – prototype conversion

Size: User	Date:	Rev: KiCad review 1
KiCad E.D.A. 10.0.6		Id: 1/1